

# Frequency Bias Signal Malfunctions

## Symptoms

- Engine does **not** respond to frequency bias request.

## How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

## Shoptalk

## Troubleshooting Summary

| STEPS   |                                                                       | SPECIFICATIONS     |
|---------|-----------------------------------------------------------------------|--------------------|
| STEP 1. | Check Customer Interface Box Wiring                                   |                    |
|         | STEP 1A. Check Generator Speed/Load Governing Bias Supply Wire        | Less than 10 ohms? |
|         | STEP 1B. Check Generator Speed/Load Governing Bias Return Wire        | Less than 10 ohms? |
|         | STEP 1C. Check Generator Speed/Load Governing Bias Signal Wire        | Less than 10 ohms? |
| STEP 2. | Check Engine Harness to Customer Interface Box Cable                  |                    |
|         | STEP 2A. Check Generator Speed/Load Governing Supply and Signal Wires | Less than 10 ohms? |
|         | STEP 2B. Check Generator Speed/Load Governing Return and Signal Wires | Less than 10 ohms? |

## STEP 1. Check Customer Interface Box Wiring

### STEP 1A. Check Generator Speed/Load Governing Bias Supply Wire

**Conditions :**

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

| Action                                                                                                                                                                                                                                                                                                                                                                        | Specification/Repair                                                                                                                                         | Next Step               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Check the generator speed/load governing bias supply wire. <ul style="list-style-type: none"><li>• Place one test lead on the generator speed/load governing bias 5 volt supply (sensor supply 4) pin in connector C3.</li><li>• Place the other test lead on the generator speed/load governing bias 5 volt supply (sensor supply 4) terminal on the X4 connector.</li></ul> | Less than 10 ohms?<br><b>YES</b>                                                                                                                             | <b>1B</b>               |
|                                                                                                                                                                                                                                                                                                                                                                               | Less than 10 ohms?<br><b>NO</b><br><b>Repair :</b><br>Replace the wire.<br>Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html). | <b>Repair complete.</b> |

### STEP 1B. Check Generator Speed/Load Governing Bias Return Wire

**Conditions :**

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|--------|----------------------|-----------|

|                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                     |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <p>Check the generator speed/load governing bias return wire.</p> <ul style="list-style-type: none"> <li>Place one test lead on the generator speed/load governing bias return (sensor return 4) pin in connector C3.</li> <li>Place the other test lead on the generator speed/load governing bias return (sensor return 4) terminal on the X4 connector.</li> </ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                         | <b>1C</b>               |
|                                                                                                                                                                                                                                                                                                                                                                       | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire.</p> <p>Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).</p> | <b>Repair complete.</b> |

## STEP 1C. Check Generator Speed/Load Governing Bias Signal Wire

### Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

| Action                                                                                                                                                                                                                                                                                                                            | Specification/Repair                                                                                                                                                                | Next Step               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <p>Check the generator speed/load governing bias signal wire.</p> <ul style="list-style-type: none"> <li>Place one test lead on the generator speed/load governing bias signal pin in connector C3.</li> <li>Place the other test lead on the generator speed/load governing bias signal terminal on the X4 connector.</li> </ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                                                                                                         | <b>2A</b>               |
|                                                                                                                                                                                                                                                                                                                                   | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the wire.</p> <p>Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).</p> | <b>Repair complete.</b> |

## STEP 2. Check Engine Harness to Customer Interface Box Cable

### STEP 2A. Check Generator Speed/Load Governing Supply and Signal Wires

**Conditions :**

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Specification/Repair                                                                        | Next Step                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------|
| <p>Check generator speed/load governing bias supply and signal wires.</p> <ul style="list-style-type: none"><li>• Place a jumper between the generator speed/load governing bias 5 volt supply (sensor supply 4) pin and the generator speed/load governing bias signal pin in the C10 connector.</li><li>• Place one test lead in the generator speed/load governing bias 5 volt supply (sensor supply 4) pin of the C3 connector.</li><li>• Place the other test lead in the generator speed/load governing bias signal pin of the C3 connector.</li></ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p>                                                 | <p><b>2B</b></p>               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the cable.</p> | <p><b>Repair complete.</b></p> |

### STEP 2B. Check Generator Speed/Load Governing Bias Return and Signal Wires

**Conditions :**

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Specification/Repair                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Next Step                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <p>Check generator speed/load governing bias supply and signal wires.</p> <ul style="list-style-type: none"><li>• Place a jumper between the generator speed/load governing bias return (sensor return 4) pin and the generator speed/load governing bias signal pin in the C10 connector.</li><li>• Place one test lead in the generator speed/load governing bias return (sensor return 4) pin of the C3 connector.</li><li>• Place the other test lead in the generator speed/load governing bias signal pin of the C3 connector.</li></ul> | <p>Less than 10 ohms?</p> <p><b>YES</b></p> <p><b>Repair :</b></p> <p>Refer to Section TF in the Troubleshooting and Repair Manual, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html), or refer to the OEM service manual for potentiometer repair instructions.</p> | <p><b>Repair complete.</b></p> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Less than 10 ohms?</p> <p><b>NO</b></p> <p><b>Repair :</b></p> <p>Replace the cable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><b>Repair complete.</b></p> |

